Customer Churn Analysis and Prediction Using Machine Learning

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***Abstract*—Customer churn prediction is a deal for phone companies because it affects how much money they make and how many customers they keep. This study looks at customer churn using the Telco Customer Churn dataset. We did some work to get the data ready looked at the data to see what it told us and used machine learning to find out what makes customers leave.**

We used a Random Forest classifier to try to predict when customers would leave. It was right about 79 percent of the time. We are not saying we got it perfect like some studies but we are trying to be realistic and see if this can really work in the real world. Customer churn prediction is what we are focusing on. We want to make sure our results are helpful, for the telecommunications industry and customer churn prediction.

***Index Terms*—Customer Churn, Machine Learning, Random Forest, Predictive Analytics, Telecommunications**

I. Introduction

When customers stop doing business with a company that is called customer churn. It is better for a company to keep the customers they already have because it is cheaper, than finding ones. The company can use machine learning to see when customer churn happens and to make a plan to keep customers so customer churn does not happen as much. Machine learning really helps with customer churn by showing what customers do before they stop being customers. It helps the company make better plans to stop customer churn.

II. Literature Review

We use models like Logistic Regression and Decision Trees and Random Forest to predict when customers will stop using our service.

Some models can seem good at predicting this but sometimes they are wrong because of a problem with the data we are using.

We have to be careful with models, like Random Forest because they can be misleading if the data is not good.

III. Methodology

A. Dataset

The Telco dataset contains 7043 customer records.

B. Data Preprocessing

So I did some data cleaning. This included dealing with missing values, which's when the data is not there and encoding categorical variables, which are the variables that have categories. I also got rid of columns that were not needed. Data cleaning is a step and it included handling missing values, in the data encoding the categorical variables in the data and removing unnecessary columns from the data.

C. Exploratory Data Analysis

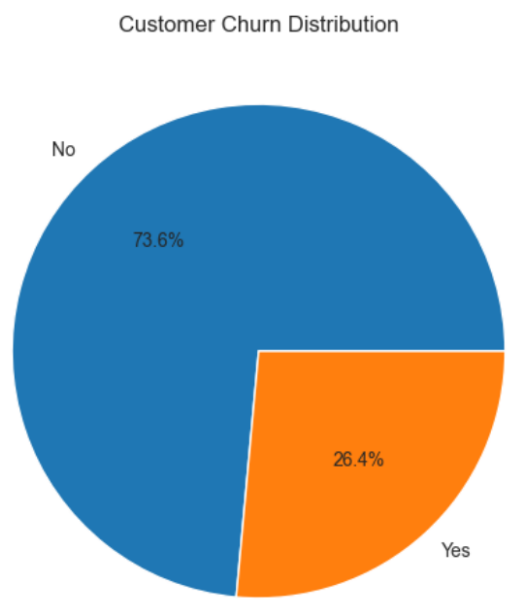

Fig. 1. Customer Churn Distribution

1) *Churn Distribution:*
2) *Contract Type vs Churn:*
3) *Payment Method vs Churn:*
4) *Internet Service vs Churn:*
5) *Tenure Distribution:*

D. Feature Engineering

We took the variables and changed them into a format that the computer can understand. Then we got rid of the features that were not needed because they were repeated.

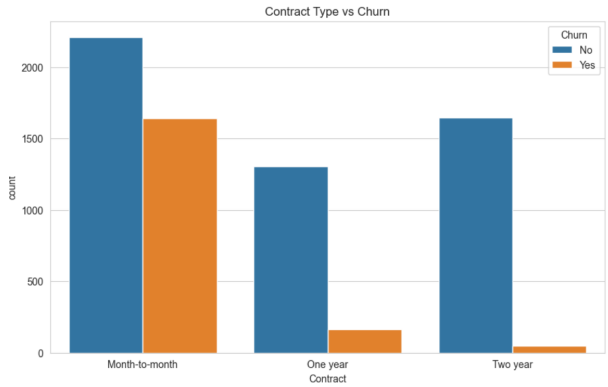

Fig. 2. Contract Type vs Churn

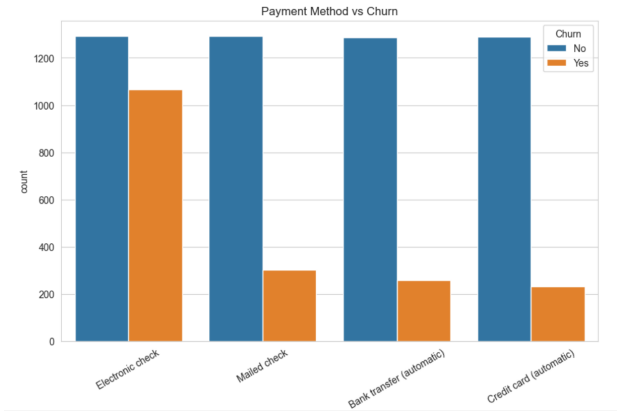

Fig. 3. Payment Method vs Churn

E. Class Imbalance

The dataset is imbalanced, affecting churn prediction.

F. Model Development

The part of making the model is really important for the system that predicts when people will stop using something. This is where we pick the tool teach it and see how well it works. For this project we used the Random Forest classifier because the Random Forest classifier is strong and the Random Forest classifier can deal with a lot of information and the Random Forest classifier does not get too good at one thing and bad, at others.

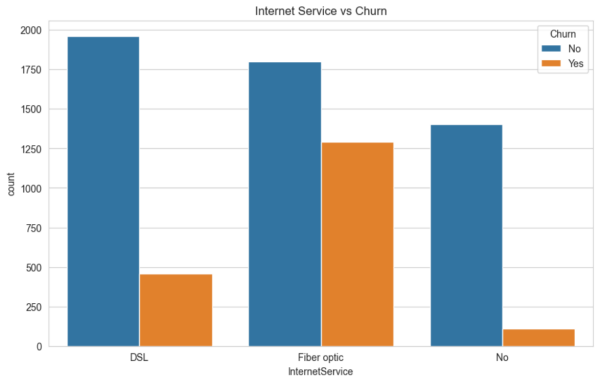

Fig. 4. Internet Service vs Churn

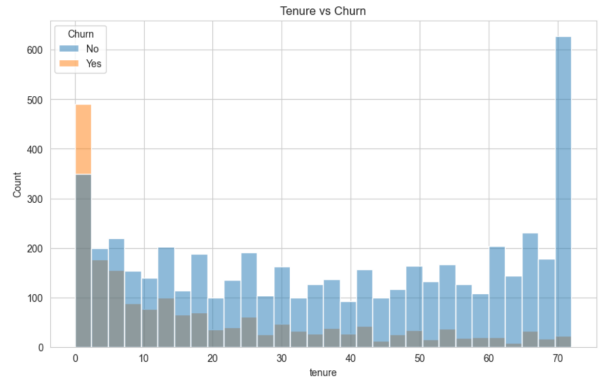

Fig. 5. Tenure Distribution

1) Model Selection: Random Forest is a technique that uses decision trees to make better predictions.

It combines the outputs of these trees to get an accurate result.

This method helps reduce errors that can happen when using one decision tree.

Random Forest works by creating trees and then averaging their predictions.

This averaging helps to lower the risk of overfitting, which's when a model is too closely fit to the training data.

By using trees Random Forest makes more reliable predictions.

Decision trees alone can be not very accurate. Random Forest improves this by using many of them.

The trees, in Random Forest are made in a way that they're not too similar which also helps in getting a better result.

2) Training Procedure: The data was split into two parts for training and testing with eighty percent for training and twenty percent, for testing. We picked the feature variables carefully to make sure we did not mess up the data.

3) Evaluation Metrics: The model was tested to see how well it worked by looking at accuracy, precision, recall and F1-score. We paid attention to recall because the classes were not balanced.

4) Challenges Faced:

- **Class Imbalance:** Most of the customers do not stop doing business with the company. This affects how well we can remember the customers who do stop the non-churn customers.
- **Data Leakage:** Initially caused unrealistically high accuracy.
- **Categorical Encoding:** Increased dimensionality.
- **Overfitting Risk:** Managed using Random Forest.
- **Feature Interpretation:** Required domain understanding.

IV. Results and Discussion

A. Performance Metrics

- Accuracy: 79%
- Precision: Moderate
- Recall: Lower for churn
- F1-score: Balanced

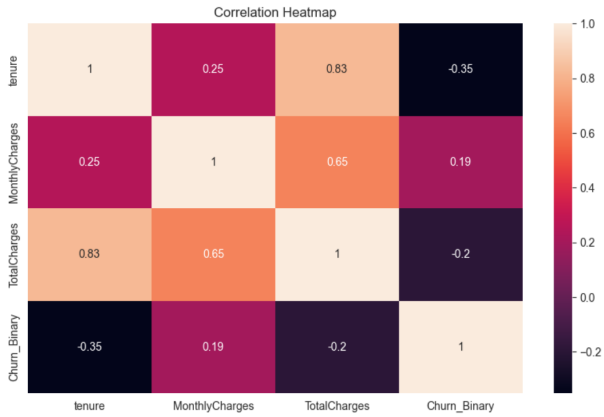

Fig. 6. Correlation Heatmap of Numerical Features

1) Correlation Analysis:

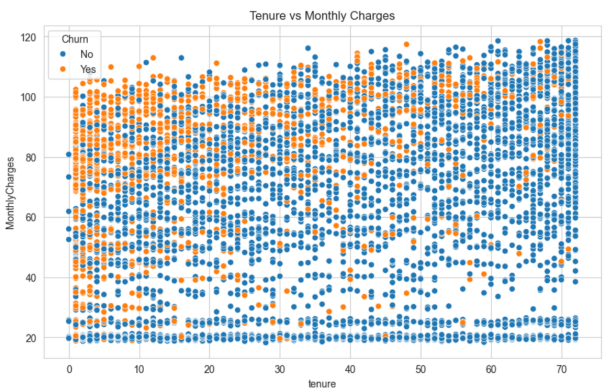

Fig. 7. Tenure vs Monthly Charges

2) Scatter Plot:

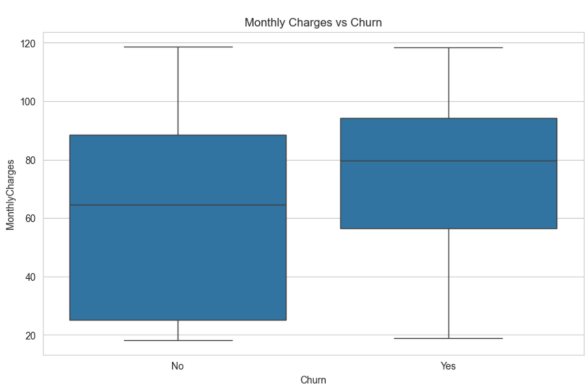

Fig. 8. Monthly Charges Distribution

3) Monthly Charges Distribution:

4) Senior Citizen vs Churn:

B. Observations

- Month-to-month contracts show higher churn
- Low tenure customers churn more
- High charges increase churn probability

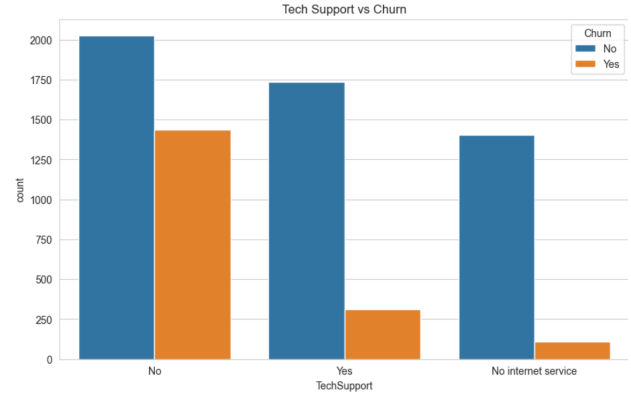

Fig. 9. Tech Support Availability vs Customer Churn

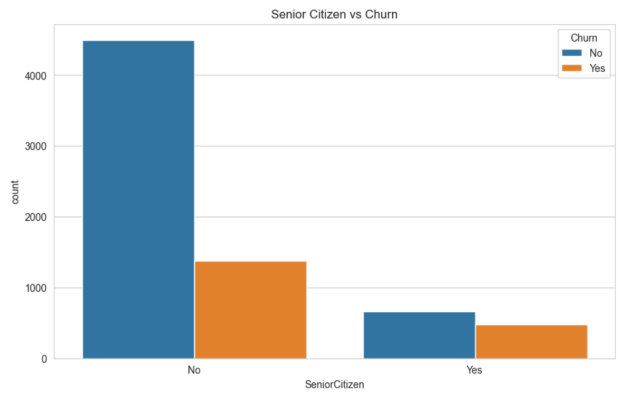

Fig. 10. Senior Citizen vs Churn

V. Business Implications

Helps find customers who might leave and makes plans to keep them.

It identifies customers who need attention.

This way you can make plans to keep them.

Helps with customer retention strategies.

VI. Limitations

Dataset is limited and imbalanced.

VII. Future Work

Use models and deal with imbalance with SMOTE.

Try models that work well and use SMOTE to balance things out.

Good models and SMOTE can help.

You should use models and SMOTE to handle the imbalance.

Advanced models are better with SMOTE, for imbalance.

VIII. Conclusion

This study looked at people who stop using a service, which is called churn. It found the main things that make people stop using the service. The study was able to figure out what makes people leave. It found the key factors that influence churn. The main things that influence churn are important to understand.

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[3] J. Han, M. Kamber, "Data Mining Concepts," Elsevier.
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